Introduction to Biomaterials

Lecture [8]

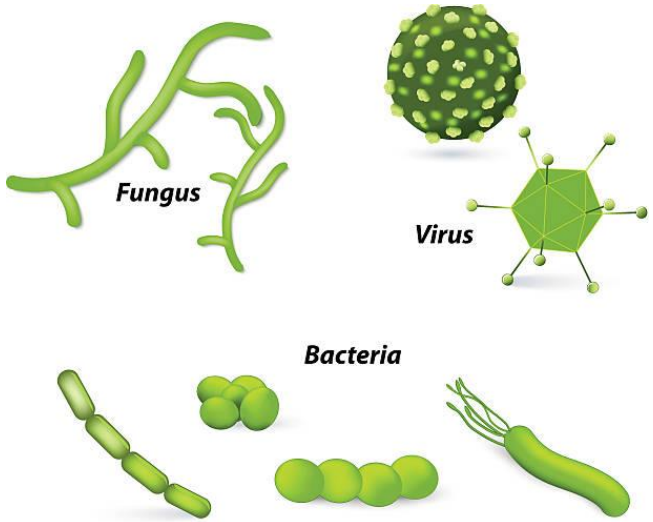
Sterilization of biomedical implants

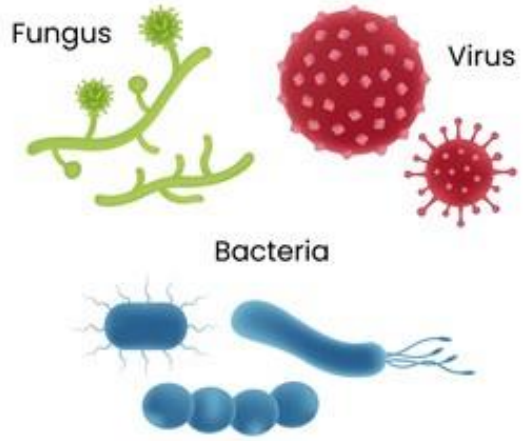
Goals

- Fundamentals and importance of sterilization.
- Different types of sterilization methods commonly used.
- Principles behind determining the type of sterilization method suitable for an application.

Why Sterilization Is Needed ?

- Microorganisms are everywhere around us
  - bacteria
  - fungi
  - viruses
- They can be found:
  - in the air
  - on object surfaces
  - on human skin





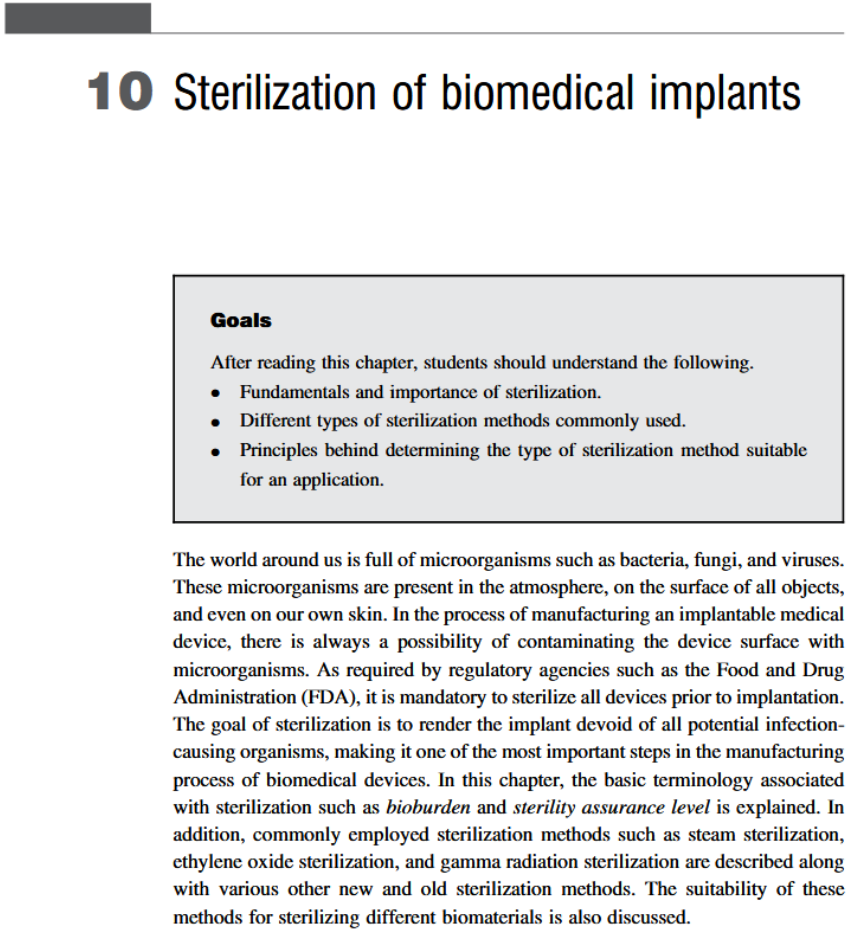
- During implant manufacturing, the device surface may become contaminated.

Main Goal of Sterilization

- **Sterilization means :** removing or destroying microorganisms from the implant.
- The **main goal** is to make the implant **free from infection-causing** organisms.
- **Regulatory agencies**, such as the **FDA**, require implantable medical devices to be sterilized before use.
- Sterilization is one of the **most important final steps** in biomedical device manufacturing.

What This Chapter Covers ?

- Basic sterilization terms:
  1. bioburden
  2. sterility assurance level, or SAL
- Common sterilization methods:
  - *steam sterilization*
  - *ethylene oxide sterilization*
  - *gamma radiation sterilization*
- Other traditional and newer sterilization techniques
- Suitability of each method for different biomaterials



10 Sterilization of biomedical implants

Goals

After reading this chapter, students should understand the following.

- Fundamentals and importance of sterilization.
- Different types of sterilization methods commonly used.
- Principles behind determining the type of sterilization method suitable for an application.

The world around us is full of microorganisms such as bacteria, fungi, and viruses. These microorganisms are present in the atmosphere, on the surface of all objects, and even on our own skin. In the process of manufacturing an implantable medical device, there is always a possibility of contaminating the device surface with microorganisms. As required by regulatory agencies such as the Food and Drug Administration (FDA), it is mandatory to sterilize all devices prior to implantation. The goal of sterilization is to render the implant devoid of all potential infection-causing organisms, making it one of the most important steps in the manufacturing process of biomedical devices. In this chapter, the basic terminology associated with sterilization such as *bioburden* and *sterility assurance level* is explained. In addition, commonly employed sterilization methods such as steam sterilization, ethylene oxide sterilization, and gamma radiation sterilization are described along with various other new and old sterilization methods. The suitability of these methods for sterilizing different biomaterials is also discussed.

Bioburden

Bio- = life / living organisms
Burden = load / amount carried

- **Bioburden** means the number of microorganisms present on the implant surface **before sterilization**.
- These microorganisms may include:
  - bacteria
  - fungi
  - viruses



- Higher bioburden means the implant needs a stronger or longer sterilization process.
- Lower bioburden means sterilization can be achieved more quickly.
- Therefore, controlling contamination during manufacturing is very important.

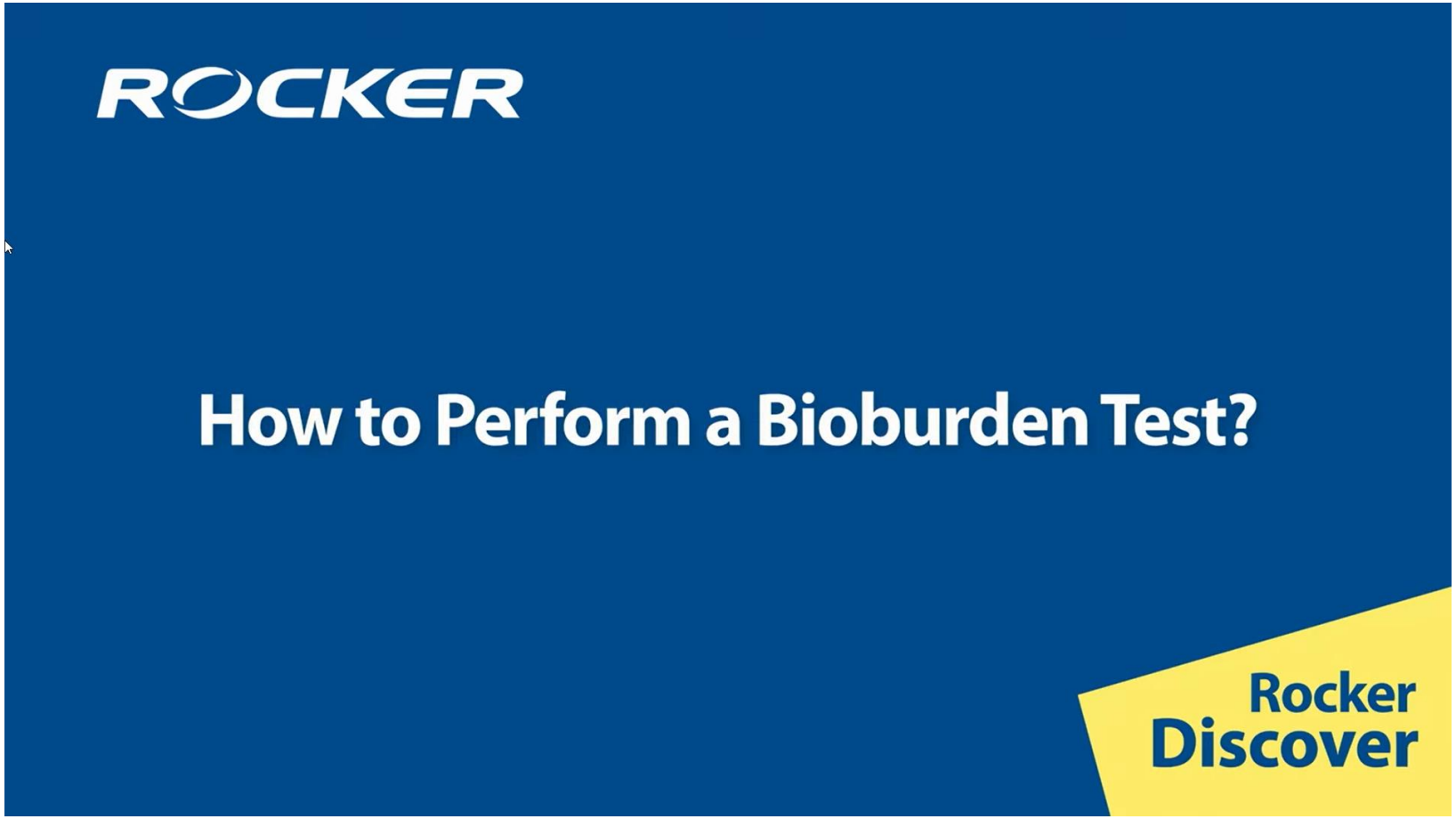
How Bioburden Is Measured ?

- **Bioburden** is **usually tested** using **random samples from production batches**.
- Around **10 to 30 product samples** may be selected.
- The samples are placed in a **sterile solution**.
- **Microorganisms** are **removed** from the **implant surface** by:
  - shaking
  - sonication
  - mechanical agitation
- The removed microorganisms are then counted using microbiological methods, such as the **standard plate method**.

How do we measure it? What is the unit?

- Wash: Rinse microbes off the device into a liquid.
- Grow: Transfer liquid to a nutrient plate and incubate.
- Count: Count the visible dots (colonies) that grow.
- CFU (Colony Forming Units) per device or per cm^2.
- The Golden Rule: Low bioburden = easy sterilization. Clean first, sterilize second.

https://www.youtube.com/watch?v=keQDFeJxjJc&t=155s



Sterility Assurance Level, SAL

- **Sterility Assurance Level, SAL**, describes the probability that one product remains unsterile after sterilization.
- For biomedical implants, the commonly accepted **SAL** is:
  - **SAL = 10^{-6}**
- This means:
  - **Only one device in one million may remain unsterile.**
- SAL is not the same as absolute zero microorganisms.
- It is a **statistical safety** level used to show that **sterilization** is **highly reliable**.
-

Sterilization Time and SAL

- To determine sterilization time, the **initial bioburden** is measured **first**.
- The implant is then **exposed** to **sterilization for different time periods**.
- As **sterilization time increases**, the **number of microorganisms decreases**.
- The **minimum** sterilization time is the **time needed** to reach **SAL = 10^{-6}**.
- Example ,**SAL** reaches **10^{-6}** after about **6 hours**.
- Usually, the actual sterilization time is kept slightly longer than the minimum time.



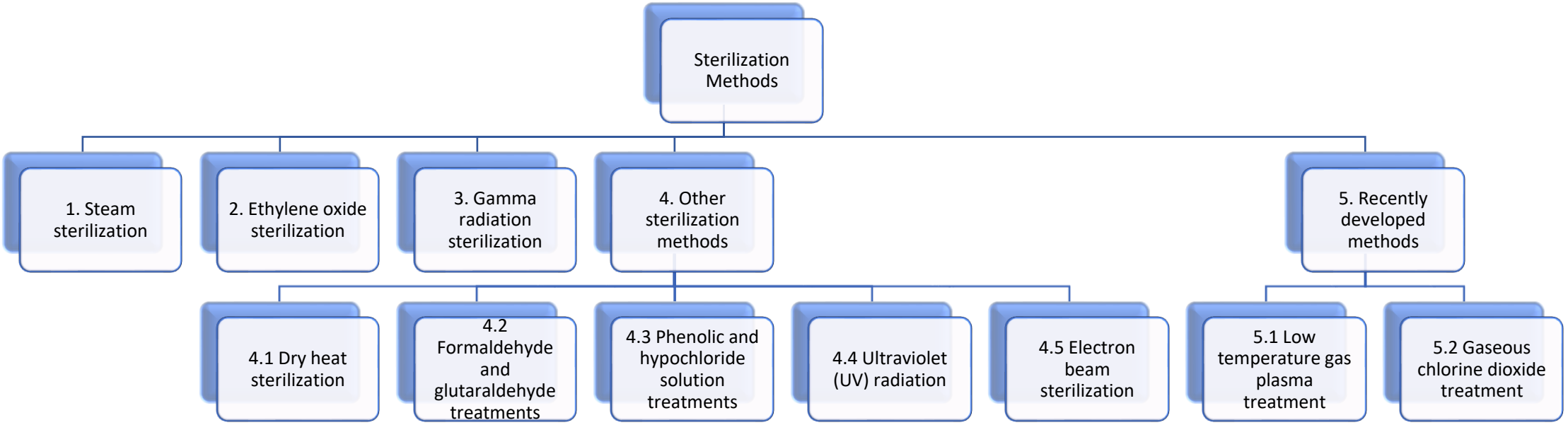
Why Extra Sterilization Time Is Needed?

- Extra sterilization time is used to avoid failure caused by natural variations. Variations may occur because:
  - different implants may have different bioburden levels
  - some microorganisms may be more resistant
  - product surfaces may not all be identical
- After sterilization, implants must be checked again.
- The aim is to confirm that sterilization did not damage:
  - surface properties
  - bulk material properties
  - mechanical behavior
  - biological performance

So;

- **Bioburden:**
  Number of microorganisms attached to the implant surface before sterilization.
- **Sterility Assurance Level, SAL:**
  Probability that one product remains unsterile after sterilization.
- **Accepted SAL for biomedical implants:**
  10^{-6} = one unsterile product in one million.
- **Important critical trade-off:**
  Sterilization must kill microorganisms without damaging the implant material.

Sterilization Methods



1. Steam Sterilization

What Is Steam Sterilization?

- **Steam sterilization** is a traditional and widely used method for sterilizing medical devices.
- It uses:
  - heated steam
  - high pressure
  - controlled exposure time



- The process is usually performed inside an **autoclave**.
- An autoclave is a **sealed chamber** that exposes devices to **pressurized saturated steam**.

Common Steam Sterilization Conditions

- Standard steam sterilization conditions:
  - **Temperature:** 121°C
  - **Pressure:** about 15 psi
  - **Exposure time:** 15 minutes
- Higher temperatures may also be used if the material can tolerate heat.
- Recommended examples:→

Temperature	Exposure Time
121°C	15 minutes
126°C	10 minutes
134°C	3 minutes

- **Higher temperature usually means shorter sterilization time.**

How Steam Kills Microorganisms?

- **Steam sterilization** mainly **kills** microorganisms by **damaging** their **essential proteins**.
- The heat and moisture cause:
  - **protein denaturation**
  - **enzyme destruction**
  - **coagulation of cellular components**
- These changes are irreversible.
- As a result, microorganisms **lose** their ability to **survive** and **reproduce**.

Important Process Condition:

- Steam must reach all surfaces of the medical device.
- The packaging material must allow steam to pass through.
- The exposure time starts only when the whole device reaches the required temperature.
- Example:
  - The 15-minute exposure begins only after the entire device reaches **121°C**.
- Poor steam contact may lead to incomplete sterilization.

Advantages of Steam Sterilization

- Simple to use
- Fast compared with many other sterilization methods
- Efficient in killing microorganisms
- Does not leave toxic chemical residues
- Suitable for many reusable surgical instruments
- Often preferred for heat-resistant medical devices



Limitations of Steam Sterilization

- Steam sterilization uses high heat and moisture.
- Therefore, it is not suitable for many heat-sensitive materials.
- It may damage or deform some polymeric biomaterials.
- It is mainly suitable for:
  - metal-based implants
  - surgical instruments
  - heat-resistant devices
- It is generally not suitable for:
  - most polymers
  - heat-sensitive biomaterials
  - moisture-sensitive materials

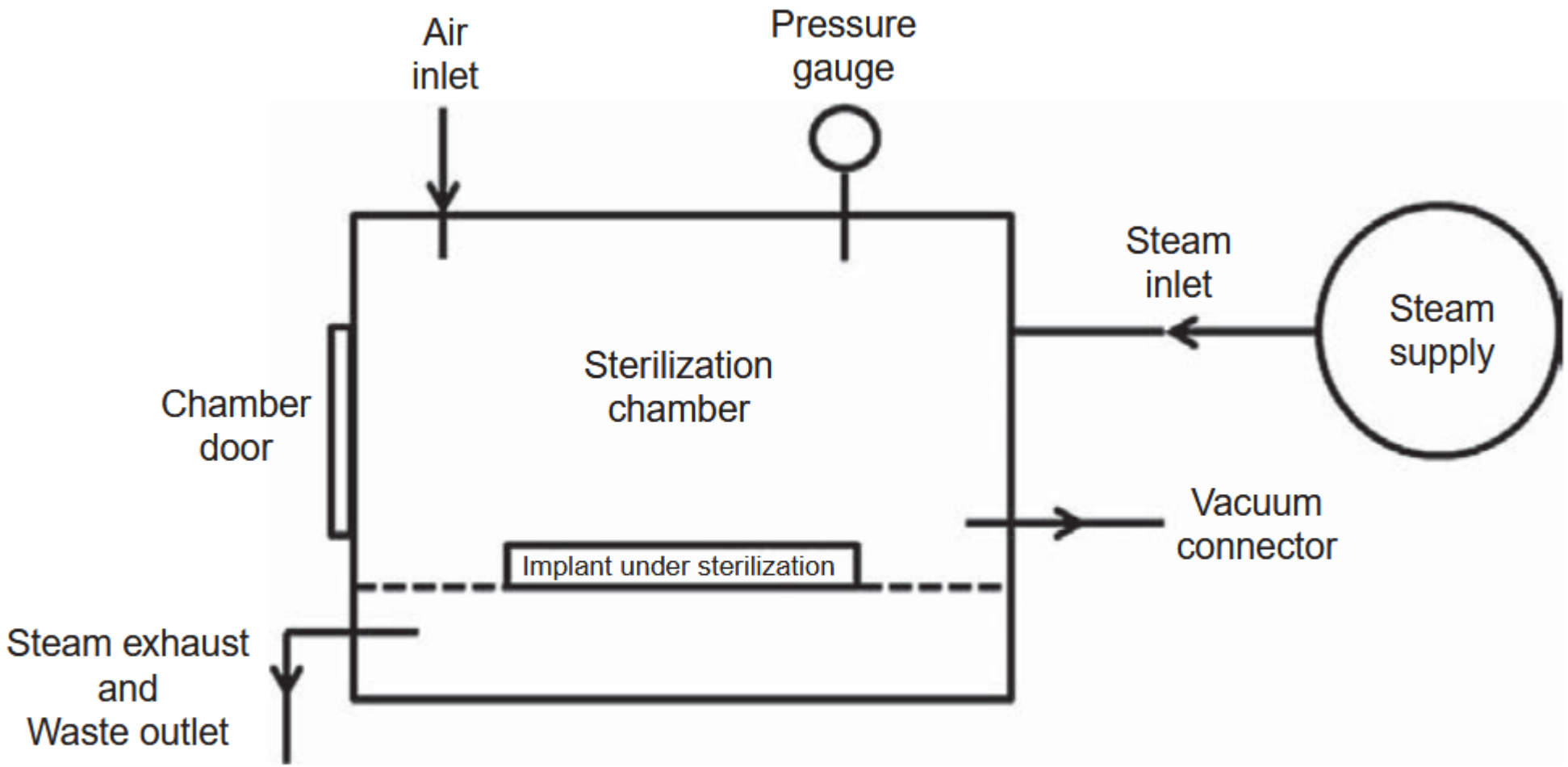


Figure 10.2

Schematic of an autoclave.

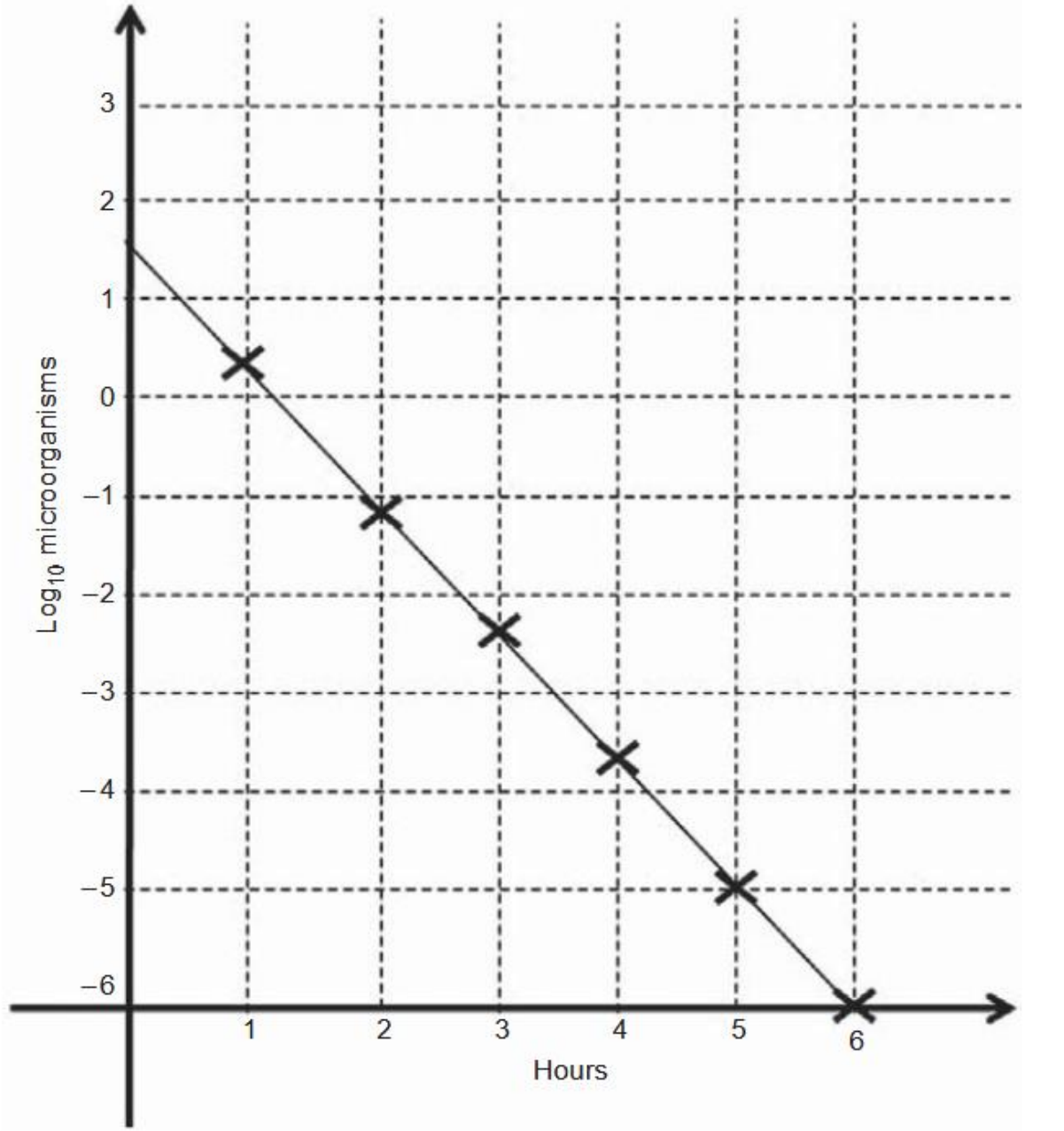


Figure 10.1

An example plot of $\log_{10}$ microorganisms versus time. SAL reaches 10^{-6} at 6 h.

2. Ethylene oxide sterilization

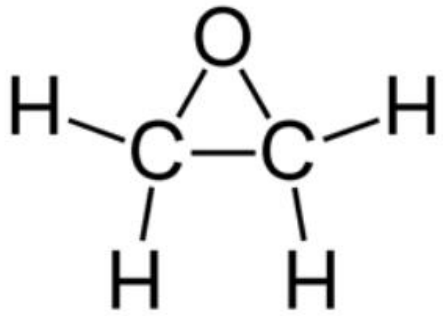
What Is Ethylene Oxide Sterilization?

- **Ethylene oxide, EtO**, is a chemical gas used for sterilization.
- It is one of the most common methods for **gas sterilization**.
- EtO sterilization is a **low-temperature process**.
- This makes it useful for materials that cannot tolerate high heat.
- It is commonly used for many polymeric and heat-sensitive biomedical devices.



Properties of Ethylene Oxide

- Ethylene oxide is:
  - colorless
  - highly reactive
  - flammable
  - toxic
  - potentially carcinogenic





- **EtO** is **rarely** used in its **pure form** because it can be **explosive**.
- It is usually **mixed** with inert gases such as:
  - nitrogen, N_2
  - carbon dioxide, CO_2
- **Because of its toxicity**, EtO sterilization requires careful **safety control**.

Basic EtO Sterilization Process

- The medical device is placed inside packaging that allows EtO gas to pass through.
- The packaged device is placed inside the sterilization chamber.
- Air is removed from the chamber using vacuum.
- Humidity is controlled using steam.
- EtO gas is then introduced into the chamber.
- After sterilization, EtO gas is removed by evacuation and flushing with sterile air or nitrogen.
- The device may also need extra aeration to remove remaining EtO traces.

Main Process Conditions

- Important EtO sterilization conditions include: **relative humidity:** 60–80%
- **EtO concentration:** about 700–800 mg/L
- **temperature:** 30–50°C
- **time:** usually 2–16 hours
- The effectiveness of EtO sterilization depends mainly on:
  - humidity
  - gas concentration
  - temperature
  - exposure time

How EtO Kills Microorganisms

- EtO **kills microorganisms** by **changing important biological molecules**.
- It mainly affects:
  - DNA
  - RNA
  - proteins
- The main chemical mechanism is called **alkylation**.
- EtO **reacts** with **functional groups** in **nucleic acids** and **proteins**.
- This **prevents microorganisms** from **surviving** and **reproducing**.

Advantages of EtO Sterilization

- EtO sterilization works at **low temperature**.
- It is suitable for **heat-sensitive** biomaterials.
- It can sterilize **complex** and **porous** structures.
- EtO gas can penetrate:
  - packaging materials
  - small spaces
  - pores
  - complex device geometries
- It is **compatible** with **many biomaterials** compared with steam sterilization.



Examples of Devices Sterilized by EtO

- EtO can be used for several biomedical implants and devices, such as:
  - vascular grafts
  - heart valves
  - intraocular lenses
  - bioabsorbable fracture fixation devices
  - polymer-based implants
  - porous biomaterials
- It is especially useful when the device **cannot tolerate** high **temperature** or **moisture.**

Limitations of EtO Sterilization

- EtO is **toxic** and requires **strict safety** control.
- Residual EtO may remain on the implant surface after sterilization.
- Toxic by-products may also form, such as **ethylene chlorohydrin**.
- **Extra aeration** may be needed after sterilization.
- The process **can take longer** than **steam sterilization.**
- Laboratory personnel must use proper protective measures to reduce EtO exposure.

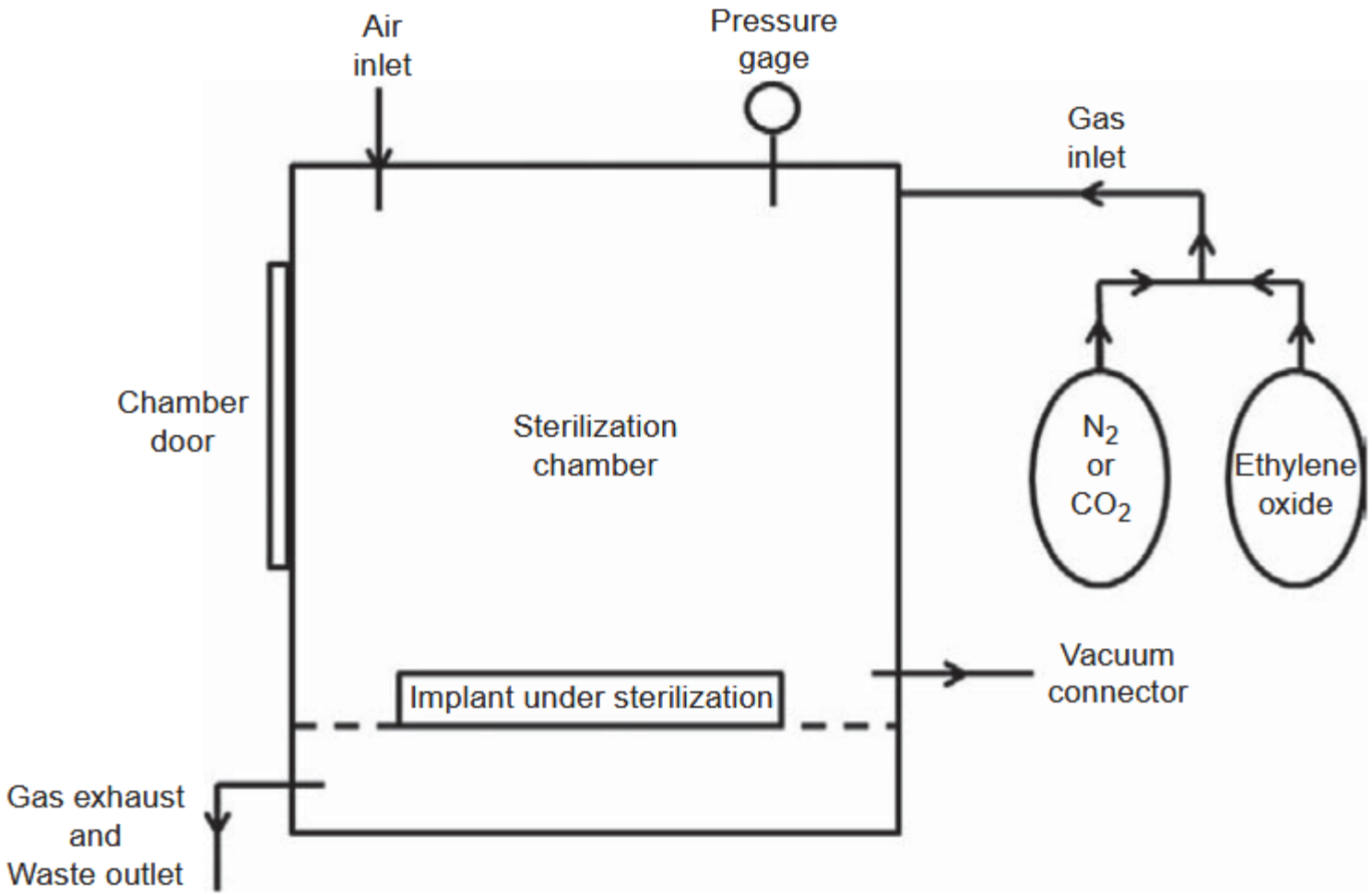


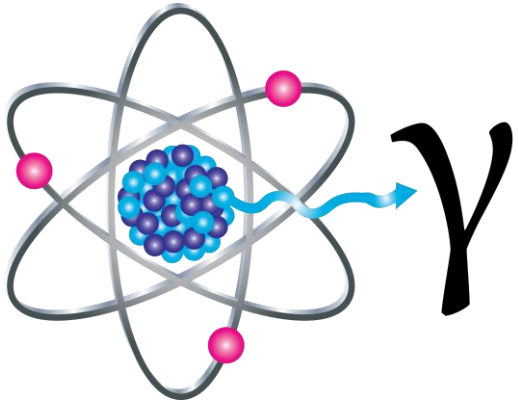
Figure 10.3

Schematic of an ethylene oxide sterilization chamber.

3. Gamma radiation sterilization

What Is Gamma Radiation Sterilization?

- **Gamma radiation sterilization** uses ionizing radiation to sterilize medical devices.
- The device is **exposed** to **gamma** rays to **kill microorganisms**.
- The most **common radiation** source is **Cobalt-60, ^{60}Co**.
- **Cobalt-60** is a radioactive isotope with a half-life of about **5.27 years**.
- Gamma sterilization is widely used for many medical materials and implantable devices.



Basic Gamma Sterilization System

- Gamma sterilization is performed inside a special radiation facility. The facility contains:
  - Radiation room
  - Thick shielding walls
  - ^{60}Co radiation source
  - Conveyor system for product movement
  - Underground water pool for source shielding when not in use
- Thick concrete walls are used to prevent gamma rays from escaping into the environment.
- The product is moved into and out of the radiation room using a conveyor system.

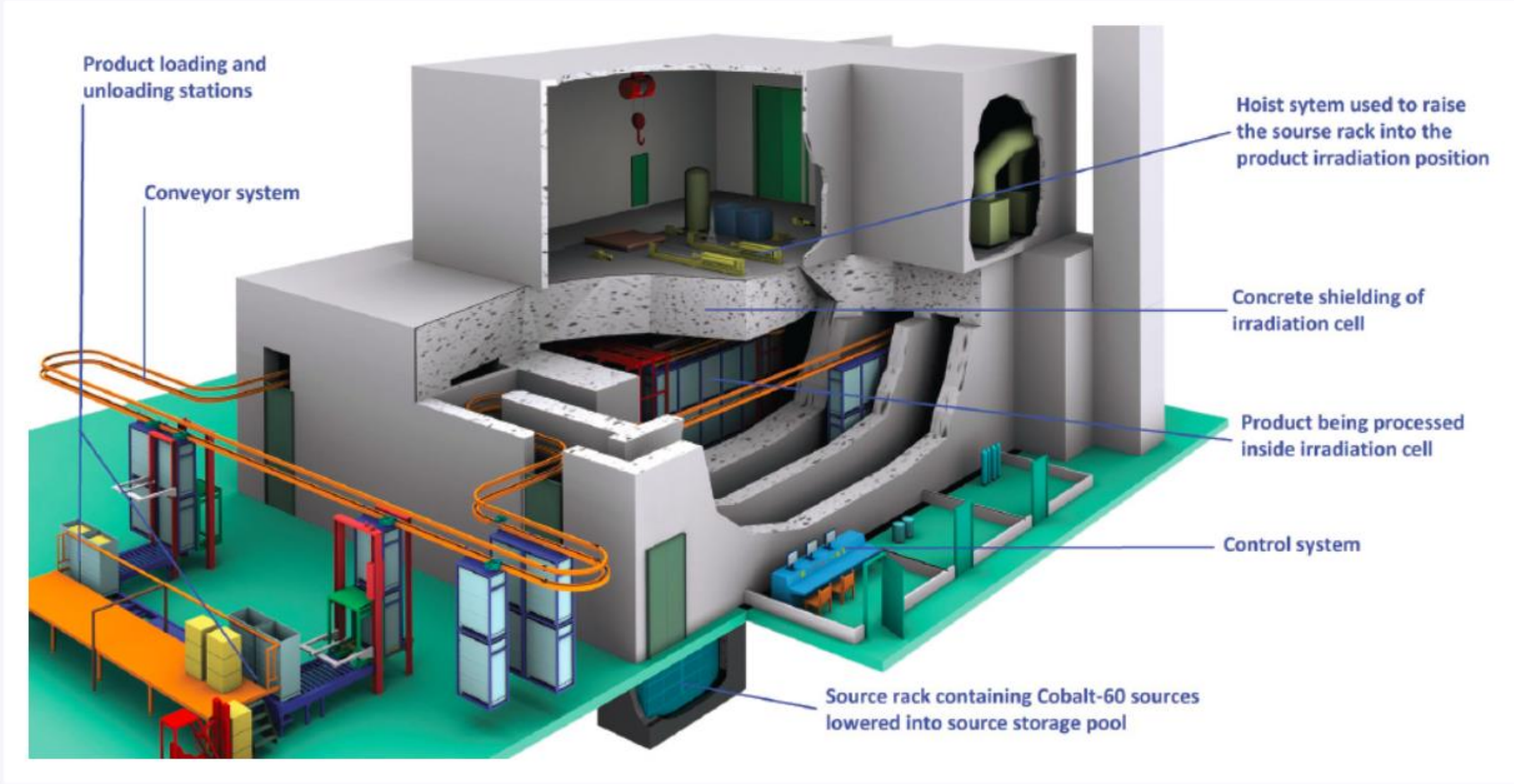


FIGURE 5.1 Typical gamma irradiator.
SOURCE: Image courtesy of SQHL Radiation Engineering Technology Co., Ltd.

Radiation Dose and Dosimeters

- A typical dose used for sterilizing medical materials is:
  - 25 kGy
- Radiation dosimeters are placed with the medical devices. Dosimeters are used to confirm that:
  - the device received enough radiation for sterilization
  - the device did not receive more radiation than the material can tolerate
- Dose control is important because too little radiation may not sterilize the product, while too much radiation may damage the material.

How Gamma Radiation Kills Microorganisms?

- Gamma radiation kills microorganisms mainly by damaging their **DNA**.
- DNA damage may happen directly or indirectly.
- Direct radiation damage can cause:
  - single-strand DNA breaks
  - double-strand DNA breaks
  - base damage
  - cross-linking between DNA strands
- Indirect damage occurs when radiation produces free radicals that attack DNA.
- When DNA is damaged, microorganisms cannot survive or reproduce.

Advantages of Gamma Sterilization

- Highly effective in killing microorganisms.
- Suitable for a wide range of medical devices.
- Can penetrate through packaging and many materials.
- Can be used for many polymers.
- Useful for products that are already sealed in final packaging.
- Compatible with several biomedical materials when the dose is properly controlled.

Polymers Commonly Compatible with Gamma Sterilization

- Several polymers can be sterilized using gamma irradiation, such as:
  - polyethylene
  - polyesters
  - polystyrene
  - polysulfone
  - polycaprolactone
- These materials generally tolerate gamma sterilization better than some other radiation-sensitive polymers.

Materials That Need Special Care

- Some polymers can be sterilized by gamma radiation, but require precautions.
- Examples include:
  - PTFE
  - polypropylene
  - ultrahigh molecular weight polyethylene, UHMWPE
  - poly(lactic acid), PLA
  - poly(glycolic acid), PGA
  - cyanoacrylates
- These materials may experience changes in their properties after radiation exposure.

Effects on Sensitive Polymers

- **PTFE and polypropylene:** highly sensitive to radiation damage
- **UHMWPE:** may undergo oxidation during gamma irradiation in air and oxidation can reduce wear resistance
- **PLA and PGA:** may undergo radiation degradation and molecular weight may decrease
- **Cyanoacrylates:** may lose elongation and tensile properties

- **Engineers must consider these changes during material and device design.**

Main Limitations of Gamma Sterilization

- Gamma sterilization facilities are expensive to build and operate.
- The process requires strict radiation safety systems.
- Some materials may degrade or lose mechanical properties.
- Radiation dose must be carefully controlled.
- Material compatibility must be checked before choosing this method.

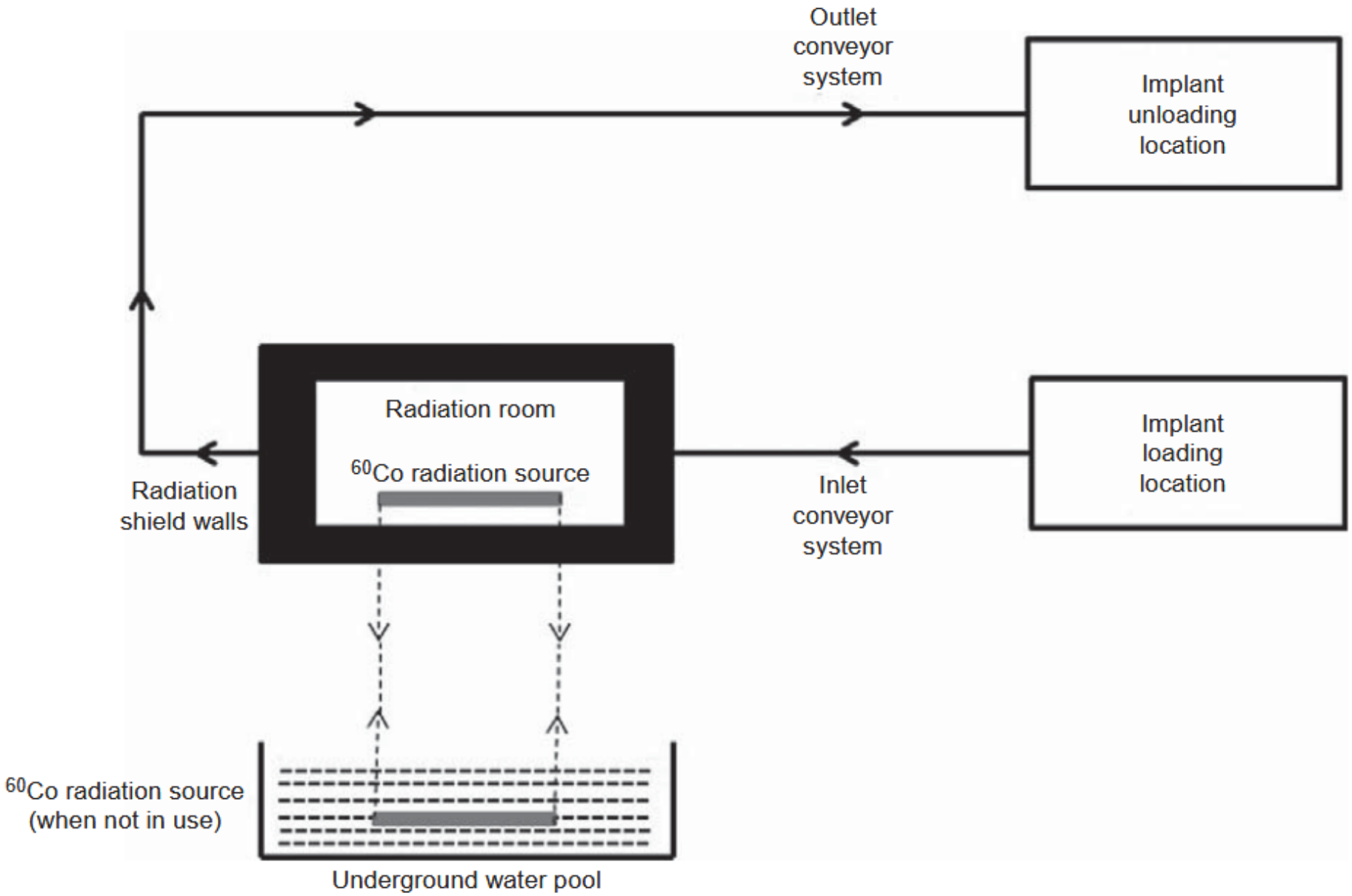


Figure 10.4

Schematic of gamma radiation sterilization.

4. Other sterilization methods

Overview of Other Sterilization Methods

- In addition to previous methods other sterilization methods may be used.
- **These include:**
    1. dry heat sterilization
    2. formaldehyde and glutaraldehyde treatments
    3. phenolic and hypochlorite solution treatments
    4. ultraviolet, UV, radiation
    5. electron beam sterilization
- These methods are generally less common because of limitations such as:
    - high temperature
    - toxicity
    - hazardous chemicals
    - poor penetration ability
    - possible material damage

4.1 Dry Heat Sterilization

- **Dry heat sterilization** uses **hot air** with **very little moisture**.
- It works at **higher temperatures** than **steam sterilization**.
- Typical conditions:→

Temperature	Exposure Time
120°C	6 hours
160°C	2 hours
180°C	30 minutes

- **Higher temperature** requires **shorter exposure** time.
- This method is mainly suitable for:
  - metallic implants
  - metal-based devices
  - heat-resistant instruments



Limitations of Dry Heat Sterilization

- Dry heat requires high temperature and long exposure time.
- It is not suitable for most polymers.
- It may damage:
  - heat-sensitive biomaterials
  - polymeric implants
  - devices with organic coatings
  - temperature-sensitive components
- Therefore, dry heat sterilization is mostly limited to metallic and highly heat-resistant materials.

4.2 Formaldehyde and Glutaraldehyde Treatments

- **Formaldehyde** and **glutaraldehyde** are **chemical sterilizing agents**.
- They can kill many types of microorganisms, including:
  - bacteria
  - fungi
  - viruses
- Their action is based on **chemical cross-linking** with:
  - proteins
  - nucleic acids
- This cross-linking inactivates microorganisms and prevents their survival.



Use and Limitations of Formaldehyde and Glutaraldehyde

- These methods may be **used** when **steam** or **ethylene oxide sterilization** is **not suitable**.
- **Glutaraldehyde** usually requires **longer immersion** time.
- They are used only when necessary because:
  - they are toxic
  - the process is labor-intensive
  - confirming the required SAL can be difficult
  - careful residue control is needed
- Their use is limited in implant sterilization compared with more common methods.

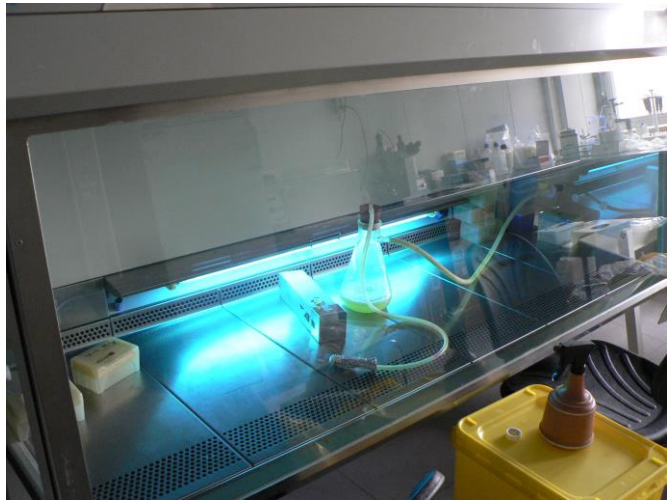
4.3 Phenolic and Hypochlorite Solution Treatments

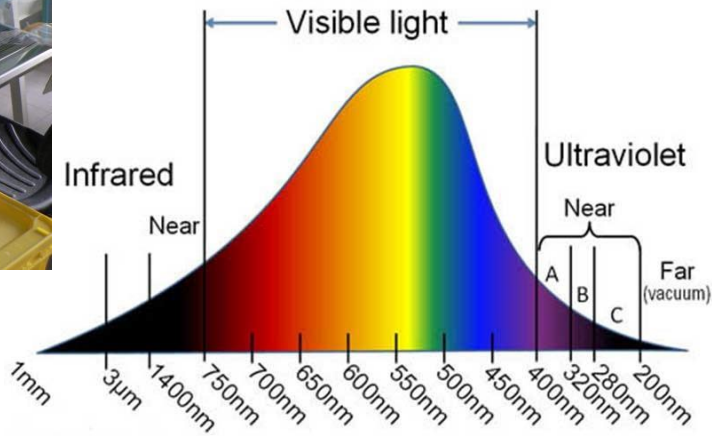
- **Phenolic compounds** and **hypochlorite** solutions can be used as **chemical treatments**.
- They are **sometimes** used for **sterilizing polymers**.
- These methods are performed at **low temperature**.
- However, they usually require **long immersion times**.
- Some **polymers** may be **damaged** during exposure.
- Their use is limited because:
  - the chemicals can be hazardous
  - the process may not be economical
  - achieving the required SAL may be uncertain



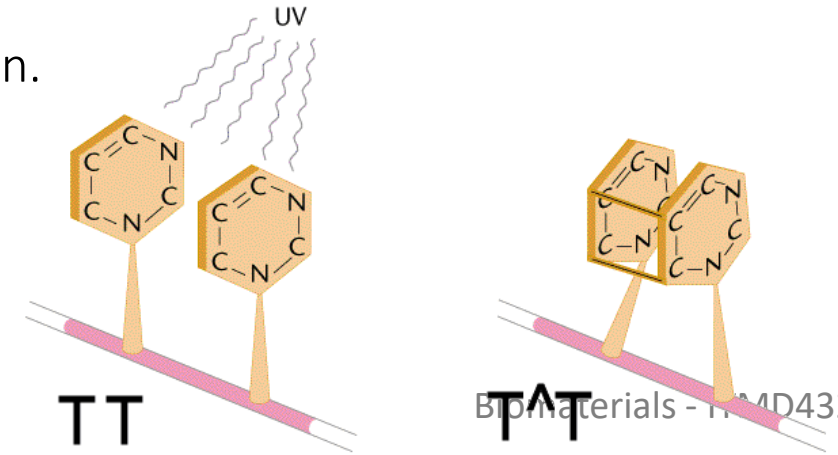


4.4 Ultraviolet, UV, Radiation Sterilization





- UV radiation is commonly used for sterilizing:
  - biological safety cabinets
  - laboratory surfaces
  - instruments used in cell culture studies
- UV radiation around **260 nm** is effective against many microorganisms.
- It damages microbial **DNA** by causing thymine dimer formation.
- This **prevents DNA replication and transcription**.
- As a result, microorganisms cannot multiply properly.

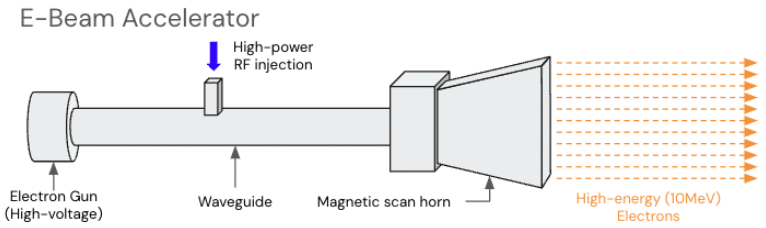


Limitations of UV Sterilization

- UV radiation is mainly effective on exposed surfaces.
- It has **poor penetration ability**.
- UV cannot effectively pass through:
  - glass
  - dirt
  - water
  - hidden or shadowed surfaces
- UV may also degrade organic molecules.
- This is a concern for:
  - polymers
  - organic coatings
  - surface-modified implants
- Therefore, UV is useful for surface sterilization, but not ideal for complex implants.

4.5 Electron Beam Sterilization

- **Electron beam** sterilization uses **accelerated electrons** to **kill microorganisms.**
- The electrons are produced using an **electron beam linear accelerator**.
- **Unlike gamma sterilization**, this method **does not use radioactive material.**
- Therefore, **no underground water pool is required**.
- The **accelerator** is **placed** inside a **shielded concrete** room **for safety**.
- A typical sterilization dose is:
  - **25 kGy**
- The exposure time may be short, around **5 minutes**.



Advantages and Limitations of Electron Beam Sterilization

- **Main advantage:** short exposure time
- **Other advantages:** no radioactive source is used
- fast processing
- useful for some medical products
- **Main limitation:** low penetration depth compared with gamma radiation
- **Penetration depth depends on:**
  - electron beam energy
  - material density
  - product thickness
- Electron beam sterilization is less suitable for thick or dense materials.

Comparison of Other Methods

Method	Main Advantage	Main Limitation
Dry heat	Simple for heat-resistant metals	High temperature
Formaldehyde / glutaraldehyde	Effective chemical action	Toxicity and residue concerns
Phenolic / hypochlorite solutions	Low-temperature treatment	Hazardous and may damage polymers
UV radiation	Good for exposed surfaces	Poor penetration
Electron beam	Very fast exposure	Limited penetration depth

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Dry heat	Simple for heat-resistant metals	High temperature
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UV radiation	Good for exposed surfaces	Poor penetration
Electron beam	Very fast exposure	Limited penetration depth

5. Recently developed methods

Recently Developed Methods

- **New sterilization methods** are being developed to **overcome limitations** of **traditional methods**.
- **Two promising methods for biomedical implants are:**
  1. low-temperature hydrogen peroxide gas plasma treatment
  2. gaseous chlorine dioxide treatment
- **These methods aim to provide:**
  - lower processing temperature
  - shorter sterilization time
  - fewer toxic residues
  - better compatibility with sensitive materials

5.1 Hydrogen Peroxide Gas Plasma Treatment

- Hydrogen peroxide gas plasma uses **H_2O_2** as the sterilizing agent.
- It was developed as an alternative to:
  - steam autoclaving
  - ethylene oxide gas sterilization
- It is considered a **low-temperature sterilization method**.
- The process is usually performed at temperatures below: **50°C**
- This makes it useful for heat-sensitive medical devices.

Advantages of H_2O_2 Gas Plasma

- **Main advantages:** low-temperature process
- Shorter processing time than ethylene oxide
- No required post-treatment aeration
- Instruments can be used soon after sterilization
- Fewer residue concerns compared with EtO
- Typical exposure time: **less than 75 minutes**
- Compared with EtO: **EtO needs about 2–16 hours**
- **H_2O_2 gas plasma is much faster**

Limitation of H_2O_2 Gas Plasma

- Hydrogen peroxide is an **oxidizing agent**.
  - This means it may react with some materials.
  - The main concern is possible changes in:
    - surface properties
    - polymer structure
    - mechanical properties
    - coating stability
    - material performance
- Therefore, implants must be tested after sterilization to confirm that the material was not damaged.

5.2 Gaseous Chlorine Dioxide Treatment

- Gaseous chlorine dioxide is another recently developed sterilization method.
- It can be used to sterilize biomedical implants.
- It is also a relatively rapid process.
- Typical exposure time: **1.5–3 hours**
- It does not require post-sterilization aeration.
- This makes it faster and simpler than some chemical sterilization methods.

Advantages of Chlorine Dioxide Treatment

- Main advantages:
  - relatively short processing time
  - no post-sterilization aeration
  - useful for biomedical implants
  - gas concentration can be monitored easily
- Chlorine dioxide gas has a green color.
- This color can be measured using a spectrophotometer.
- Measuring the gas concentration helps control the sterilization process more accurately.

Limitation of Chlorine Dioxide Treatment

- Chlorine dioxide is also an **oxidizing agent**.
- Because of this, it may affect some biomaterials.
- Careful monitoring is needed to make sure sterilization does not change:
    - surface properties
    - bulk material properties
    - mechanical strength
    - chemical stability
    - biological performance
- Material compatibility must be confirmed before using this method for implants.

Criteria for an Ideal Sterilization Method

- An ideal sterilization method should be:
  - low temperature
  - fast
  - effective against microorganisms
  - able to penetrate the full implant
  - safe for users and patients
  - economical to set up and operate
- It should not:
  - leave toxic residues
  - produce harmful by-products
  - cause oxidation
  - damage surface properties
  - change bulk material properties

Think with me 🤔🤔🤔

- List all the sterilization methods mentioned in the lecture, organize them in a tree structure, and briefly explain the nature of each method: heat, gas, chemical, or radiation. For radiation methods, specify the type of radiation used.



Thank you